```
incumbents = torch.zeros((num_experiments, num_iters))

pbar = tqdm(total=num_experiments * num_iters)

for exp in range(num_experiments):
    torch.manual_seed(exp) ❶
    train_x = bounds[0] + (bounds[1] - bounds[0]) * torch.rand(1, 4) ❶
    train_y = flight_utility(train_x) ❶

    for i in range(num_iters):
        incumbents[exp, i] = train_y.max() ❷

        model, likelihood = fit_gp_model(train_x, train_y) ❷

        ... ❸

        next_y = flight_utility(next_x) ❹

        train_x = torch.cat([train_x, next_x]) ❹
        train_y = torch.cat([train_y, next_y]) ❹

        pbar.update()
```

❶ Randomly initialize the training data.

❷ Keep track of optimization progress and update the predictive model.

❸ Define the policy and find the next batch to query.

❹ Query the points recommended by the policy and update the training data.

For the Probability of Improvement policy, we use the following code:

```
policy = botorch.acquisition.monte_carlo.qProbabilityOfImprovement(
    model, best_f=train_y.max()
)
```

For the Expected Improvement policy, we use the following code:

```
policy = botorch.acquisition.monte_carlo.qExpectedImprovement(
    model, best_f=train_y.max()
)
```

For the Upper Confidence Bound policy, we use the following code:

```
policy = botorch.acquisition.monte_carlo.qUpperConfidenceBound(
    model, beta=2
)
```

These three policies can then be optimized using the helper function `optimize_acqf()` as follows:

```
next_x, acq_val = botorch.optim.optimize_acqf(
    policy,
    bounds=bounds,
    q=batch_size,
    num_restarts=100,
    raw_samples=200,
)
```

Otherwise, for either Thompson sampling or Max-value Entropy Search, we first need to define the Sobol sequence:

```
sobol = torch.quasirandom.SobolEngine(4, scramble=True) ❶
candidate_x = sobol.draw(5000)
candidate_x = (bounds[1] - bounds[0]) * candidate_x + bounds[0]
```

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